

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

# ICT Class 8 - Formative Assessment 1

Formative Assessment (FA-1) - Answer Key

Class: Class 8

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Functions

ICT Teaching Resources Portal

## Answer Key - Grading Guidelines

**Note to Teacher:** This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

### Part A: Written Concept Paper (20 Marks)

#### Section I: Multiple Choice Questions (5 x 1 = 5 marks)

| Q.No | Answer | Explanation                                                                     |
|------|--------|---------------------------------------------------------------------------------|
| 1    | a      | The function returns "Hello, " + name, so greet("Alice") returns "Hello, Alice" |
| 2    | c      | The def keyword is used to define a function in Python                          |
| 3    | b      | A pure function is one that returns a value and has no side effects             |
| 4    | c      | Python functions return None by default if no return statement is present       |
| 5    | a      | Parentheses ( ) are used to call a function in Python                           |

#### Section II: Short Answer Questions (5 x 2 = 10 marks)

1. **Define a function. Give its syntax in Python.**

- **Answer:** A function is a reusable block of code that performs a specific task. Syntax: `def function_name(parameters):` followed by an indented block of code.
- **Marking:** 1 mark for correct definition, 1 mark for correct syntax.

2. **Differentiate between arguments and parameters with one example each.**

- **Answer:** Parameters are variables listed in the function definition (e.g., `def greet(name):` where `name` is a parameter). Arguments are actual values passed when calling the function (e.g., `greet("Alice")` where `"Alice"` is an argument).
- **Marking:** 1 mark for explaining parameters with example, 1 mark for explaining arguments with example.

3. **Write the output of the factorial function.**

- **Answer:** 120
- **Marking:** 2 marks for correct answer ( $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ ).

4. **Explain the difference between a local variable and a global variable inside a function.**

- **Answer:** A local variable is defined inside a function and can only be accessed within that function. A global variable is defined outside any function and can be accessed throughout the program. To modify a global variable inside a function, the `global` keyword must be used.
- **Marking:** 1 mark for explaining local variable, 1 mark for explaining global variable.

5. **What is the purpose of the return statement in a function? Give an example.**

- **Answer:** The return statement sends a value back to the caller of the function. Example: `def add(a, b): return a + b` – when called as `add(3, 5)`, it returns 8.
- **Marking:** 1 mark for explaining purpose, 1 mark for correct example.

#### Section III: Long Answer Questions (5 x 1 = 5 marks)

1. **Write a Python function `is_prime(n)` that checks whether a given number `n` is a prime number or not.**

- **Answer:**

```
def is_prime(n):
    """Check if a number is prime."""
    if n <= 1:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

# Test with 5 values
for num in [2, 3, 4, 17, 20]:
    print(f"{num} is prime: {is_prime(num)}")
```

- **Marking:** 1 mark for function definition, 1 mark for docstring and edge cases, 1 mark for correct logic, 1 mark for test calls, 1 mark for output display.

## Part B: Hands-on Practical Lab Task (5 marks)

### BMI Calculator Program:

- **Answer:**

```
def calculate_bmi(weight_kg, height_m):
    """Calculate BMI with input validation."""
    if weight_kg <= 0 or height_m <= 0:
        return "Invalid input: weight and height must be positive"
    bmi = weight_kg / (height_m ** 2)
    if bmi < 18.5:
        category = "Underweight"
    elif bmi < 25:
        category = "Normal"
    elif bmi < 30:
        category = "Overweight"
    else:
        category = "Obese"
    return bmi, category

# Test with 4 persons
persons = [
    ("Alice", 55, 1.65),
    ("Bob", 85, 1.75),
    ("Charlie", 45, 1.60),
    ("Diana", 100, 1.70)
]

print(f"{'Name':<10} {'BMI':<8} {'Category':<12}")
print("-" * 30)
for name, weight, height in persons:
    bmi, category = calculate_bmi(weight, height)
    print(f"{'Name':<10} {'bmi':<8.1f} {'category':<12}")
```

- **Marking:** 1 mark for function definition, 1 mark for input validation, 1 mark for BMI categorization, 1 mark for function calls, 1 mark for formatted output.

## Part C: Notebook Maintenance (5 marks)

- **Neatness and organisation of notebook (2 marks):** Check for clean, organized notebook.
- **Completion of all assigned classwork and homework (2 marks):** Verify all work is complete.
- **Proper headings, dates, and indexing (1 mark):** Ensure proper formatting throughout.